S2 E10

BK

Season 2: Retrieval and RAG

Grounding and citations.

An answer can sound sourced and still not be. Ask the model to cite each claim, then check every citation in code. Tested against the 520 test questions.

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Applied AI

S2 E10

Why it matters

A right answer, with no way to check it

1	2	3	4	5	6	7	8	9	10	11
Load	Chunk	Embed	Rewrite	Search	Filter	Rerank	**Prompt**	Answer	**Cite**	Score

HOW OUR SYSTEM ANSWERS A QUESTION, SO FAR

Find
search, filter and rerank pick the best 5 passages

→

Prompt
the 5 passages and a plain instruction go to the model

→

The model
answers in its own words

A REAL TEST QUESTION

What does error E-2194 mean? Five passages in the prompt

The model answered correctly: a factory reset on the Ashby tumble dryer was interrupted. But which of the five passages did it use? The answer does not say.

RIGHT ANSWERS, OF 400 ANSWERABLE

359

ANSWERS THAT NAMED A PASSAGE

0

This episode

Can the model point to the passage behind each claim, and can code check that the passage really says it?

Where stages 8 and 10 sit

Two stages change together this time. Both run live, on every question.

8. Prompt
a new instruction: cite each claim

→

9. Answer
the model writes [P1], [P2] after each claim

→

10. Cite
code reads each marker and checks it

Until now	Stage 8 used the plain instruction. Stage 10 did nothing.
Why both	Stage 10 reads markers that only the new instruction asks for. Neither means anything alone.

What "grounded" means

Answering only from the passages given, not from what the model learned in training. A citation is how an answer shows it, and a check is how you know.

Live, every question

The new instruction, sentence by sentence

Stage 8 swaps the plain instruction for three sentences. Each does a different job.

1 `Answer the question using only the passages below.`
Grounding: the passages, and nothing else.

2 `After each claim, cite the passage it comes from, as [P1], [P2] and so on.`
Citations: a pointer after every claim, to a numbered passage.

3 `If the passages do not contain the answer, reply that you could not find it.`
E9’s “not found” sentence, word for word.

Watch sentence 3

E9 showed this sentence alone lifts "not found" replies. Some of this episode’s results come from it, not from citations (slide 13).

S2 E10

Mental model

A citation is a pointer, not proof

The E-2194 answer again, now under the new instruction. The marker points to passage 1.

THE ANSWER

"E-2194 means the Ashby tumble dryer's factory reset was interrupted before its saved settings could be cleared, so your previous preferences may still be active. [P1]"

PASSAGE 1

Resetting your Ashby tumble dryer

"On the Ashby tumble dryer, E-2194 means a factory reset was interrupted before the saved settings could be cleared. Your previous preferences may still be active."

The risk

S1 E3 warned of a new failure: a confident answer with a citation attached, where the passage does not say it.

So the pattern is

Ask for citations, then check them. A citation nobody checks only makes a wrong answer look trustworthy.

Live, every question

Stage 10: Cite

WHAT CHANGES IN THIS STAGE

Code reads every [Pn] marker, finds that passage in the prompt, and checks it holds the evidence for the question. No model call, so it costs nothing.

```
E9  cite: {mode: passthrough}
```

```
E10 cite: {mode: parse}
```

episodes/s2-e10-grounding-and-citations/analyse.py — the whole function

```python
def cited(answer, n_passages):
    """The passage numbers an answer cites, from markers like
    [P2] or [P2-P3]; numbers outside the prompt are ignored."""
    found = set()
    for first, last in MARKER.findall(answer):
        lo, hi = int(first), int(last or first)
        found.update(range(lo, hi + 1))
    return sorted(n for n in found if 1 ⩽ n ⩽ n_passages)
```

Supported	The cited passage holds the labelled evidence for the question, by the same rule as a search hit.

Worked example

One answer, two claims, two checks

A TWO-PART QUESTION FROM THE TEST

How long should I keep my proof of purchase for a Tessa dishwasher, and how much would it cost to have it collected if I return it? Answer: 62 months, and £27.00

DOES THE CITED PASSAGE HOLD WHAT THE CLAIM SAYS?

Claim 1	"Keep your proof of purchase for any future warranty claim." [P1] P1, the Tessa warranty article, says nothing about 62 months.	no
Claim 2	"A Tessa dishwasher collection for a change-of-mind return costs £27.00..." [P2] P2, the Tessa returns policy: "Collection of a returned Tessa dishwasher costs £27.00."	yes

What the check shows

Exactly which half to distrust. The 62 months was never in search's top 50, so the model could not cite it. Without citations, the answer reads as one confident whole.

The test: two versions of the system

Both ran end to end on all 520 test questions. Nothing else in the system changed.

1 **Plain** the reference
The system after E9: the plain instruction, and stage 10 doing nothing.

2 **Grounded and citing**
The three-sentence instruction at stage 8, and stage 10 reading the markers.

The rule to join, set in advance

The model must cite at least 90% of the time, and neither right answers nor handled right may be clearly worse than the reference's.

Why a lower bar than E9?

E9 changed how well the system answers, so it had to clearly beat the reference. Citations are a new ability, so here it is enough to add them without making answers worse.

What is measured

The season's usual scores, plus three new ones for citations, all counted in code.

NEW: THE CITATIONS

Citing	Answers, other than "not found", with at least one [Pn] marker.
Supported	Citations whose passage holds the evidence for the question.
Right and cited	Right answers that cite the passage holding the evidence.

AS BEFORE

Right answers	Of the 400 answerable questions, judged right by a judge model.
Said not found	Of the 120 the library cannot answer.

Finding

Asked to cite, the model cited every time

Round 2, the grounded and citing system, all 520 test questions.

PLAIN INSTRUCTION

0 of 400

answers given named a passage. This model never cites unasked.

GROUNDED AND CITING

365 of 365

answers cited at least one passage.

Supported	365 of 400 citations pointed to a passage holding the evidence: 91%.
Right and cited	Of 346 answers judged right, 343 cited the passage that holds the answer.

365 answers, not 400: the new instruction also says "not found" more often (slide 13).

What the 35 flagged citations really were

35 of 400 citations pointed to a passage without the labelled evidence. Flagged is not the same as false.

Beside a good one	Most sit next to a supported citation, on a second point that the other passage makes.
Another true source	"What should I do when my washer flashes an error code?" was answered, rightly, from the flashing-light article. The labels mark a different article.
Nothing supported	Only 5 answers cited flagged passages alone.

The lesson

A check by evidence labels is free and exact, but cannot read meaning. A judge call per citation could, at a cost. Use flags as a list to read, not a verdict.

Finding

Citations came at no measured cost

Round 2: grounded and citing against plain, the whole system, all 520 test questions.

	RIGHT ANSWERS	SAID NOT FOUND	HANDLED RIGHT
Plain the reference	**359** of 400	**83** of 120	**442** of 520
Grounded and citing	**354** -5	**107** +24	**461** +19

Answerable	Of the 400: 8 gained, 13 lost. A split like that happens by chance about 4 times in 10.
No answer to find	Of the 120: 25 more now say "not found", 1 no longer does.
Faithful	By the judge, 351 of 352 answers against 359 of 363. Both near the ceiling.

Finding

The “not found” gain is mostly E9’s sentence

Both runs started from the same reference, which said "not found" 83 times in 120.

E9: THE SENTENCE ALONE

105 of 120

"If the passages do not contain the answer, reply that you could not find it."

E10: ALL THREE SENTENCES

107 of 120

The same sentence, plus grounding and citations.

The big jump, from 83, came from sentence 3. Grounding and citations add 2 more, which could be luck.

Next episode

E11 separates the two properly, using E9’s run, and asks whether the citation rule changes the answers themselves.

The decision

It joins: citations, and answers no worse

The rule had three parts, set before the run. All three held.

Cites	At least 90% of answers cite a passage. It was 365 of 365.	yes
Right answers	Not clearly worse: 354 against 359, and the likely ranges overlap.	yes
Handled right	Not clearly worse: 461 of 520 against 442, so higher.	yes

What joins

The three-sentence instruction and the citation check. E9’s "not found" sentence joins with them. From here, every answer cites its passages, and says "not found" when it should.

What this does not prove

- **Support by labels only.** A passage that says the same thing in other words counts as unsupported.
- **No judge call per citation.** A citation to the right passage that misstates it is caught only by the judge’s faithfulness verdict.
- **Two changes at once.** "Not found" and citations arrive together. E11 separates them.
- **The judge** is gemini-3.6-flash, checked against E3’s sample.

For engineers

Run it yourself

Both full runs, and the citation check.

```
# rebuild every result, chart and citation check: no key, no cost
$ uv run lab rag analyse episodes/s2-e10-grounding-and-citations/config.yaml

# the full run (keys needed)
$ uv run lab rag run episodes/s2-e10-grounding-and-citations/config.yaml
```

The check	`cited()` and `citation_report()` in analyse.py. No model call.
Cost	$1.36 for the episode: $0.15 of answering, $1.21 of judging.

For leaders

Ask where each answer came from

Here, one instruction made every answer point to its source, and code checked all 400 citations for free.

Ask your team

Can our assistant show which document each claim came from?

Ask your team

Does anyone check that the cited document actually says it, or do we just display the link?

Ask your team

When a citation points to the wrong document, would we ever know?

Scoreboard

Stages 8 and 10 join the reference system

1 Load	2 **Chunk**	3 Embed	4 Rewrite	5 Search	6 **Filter**	7 **Rerank**	8 **Prompt**	9 Answer	10 **Cite**	11 Score

	SEARCH	RIGHT	NOT FOUND	LEAKS
After E9 `plain instruction`	**344** of 400	**359** of 400	**83** of 120	**0** of 480
After E10 `grounded, citing`	**344** of 400	**354** of 400	**107** of 120	**0** of 480

> Lime: joined. Coral: tested, not joined. Every later episode starts from the bottom row, and every answer now carries checkable citations.

Decision rule

Require every claim to cite a passage, then check that the passage supports it.

Next: S2 E11, a field note

Does asking for citations change the answers themselves? E11 separates the citation rule from the "not found" sentence.

Where this fits

The series map

S1 LLMs	**S2** Retrieval	**S3** Strategy
S4 Adapting	**S5** Agents	**S6** Economics
S7 Data	**S8** Evaluation	**S9** AI teams
S10 Reliability	**S11** Security	**S12** Delivery
S13 Performance	**S14** Resilience	**S15** Governance

☐ Leadership season

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Episode guide brendankellyai.github.io

Code github.com/BrendanKellyAI/applied-ai-lab